Plotluck

Where could your restaurant work?

Describe the restaurant you want to open. Plotluck ranks the English postcode districts where a place like it has done well with diners.



IBM Z Datathon 2026 · Community Impact

Team Jellyfishes.AI

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Choosing where to open is a costly guess

For a small restaurant, the wrong location can sink the business. It is hard to compare areas before signing a lease, and a first-time owner rarely has more than gut feel and a few site visits.

Plotluck helps founders compare English postcode districts for the restaurant they want to open. It is built for independent founders, including people bringing under-represented cuisines to new areas.

> One of our teammates watched his father’s restaurant close during COVID. We wanted to make the first location search a little less guesswork.

Plotluck is free, runs in the browser and needs no account.

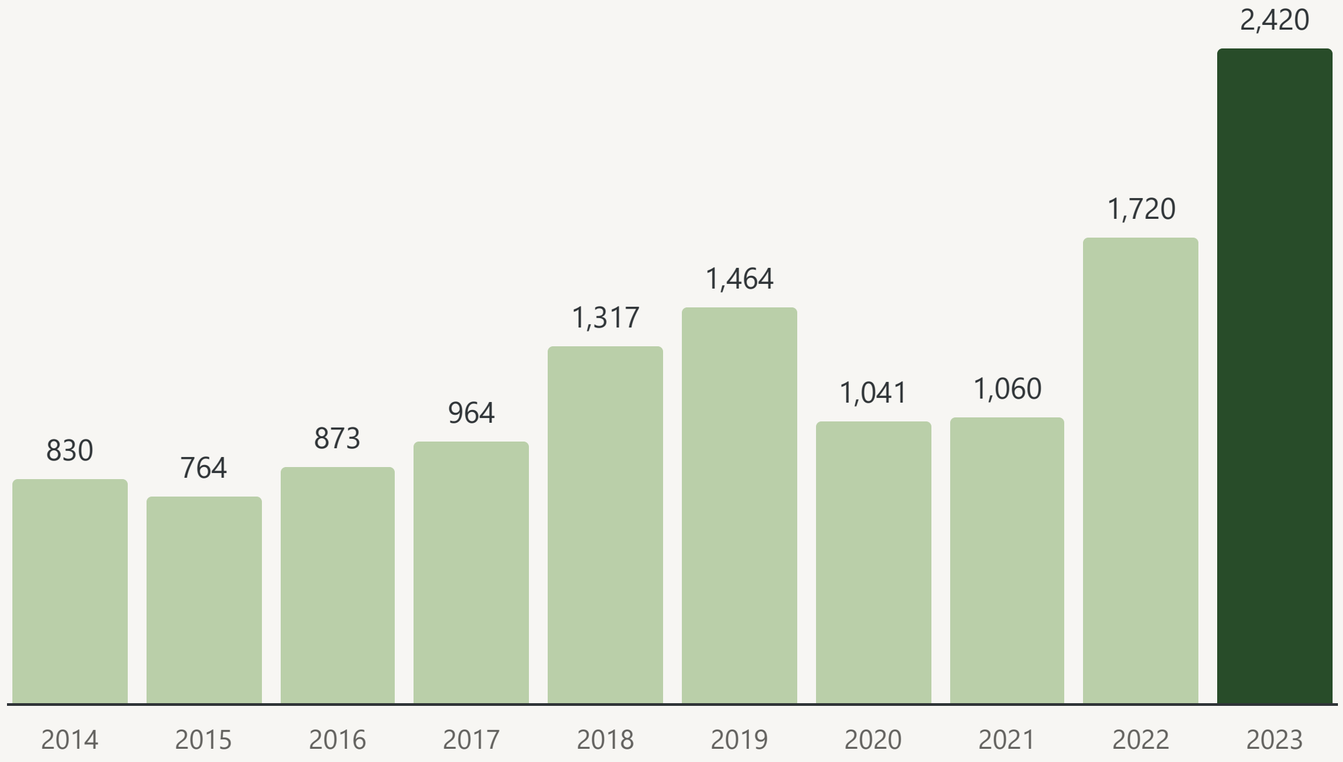
Will people here want this cuisine?

Is there already too much of it nearby?

Can the area carry my prices and rent?

Restaurant insolvencies nearly tripled in ten years

Company insolvencies, restaurants and mobile food service, England and Wales



2,420

restaurant and food-stand companies became insolvent in 2023, about **seven every day**, up from 830 in 2014.

1 in 7

company insolvencies in 2023 were food and drink service businesses: 3,474 of 25,163.

These count registered companies only. Sole traders and partnerships that close are not included.

Source: Insolvency Service, *Company Insolvency Statistics, December 2024*, industry tables A1a and A1b (SIC 2007 group 561 and division 56), England and Wales. gov.uk/government/statistics/company-insolvencies-december-2024

One description in, a ranked district shortlist out

One description starts the search. Plotluck asks only when a detail is missing or unclear.

Describe

Tell Plotluck what you want to open and where you want to look, in your own words.

→

Clarify

If a detail is unclear, answer a focused follow-up question. Details already read are kept.

→

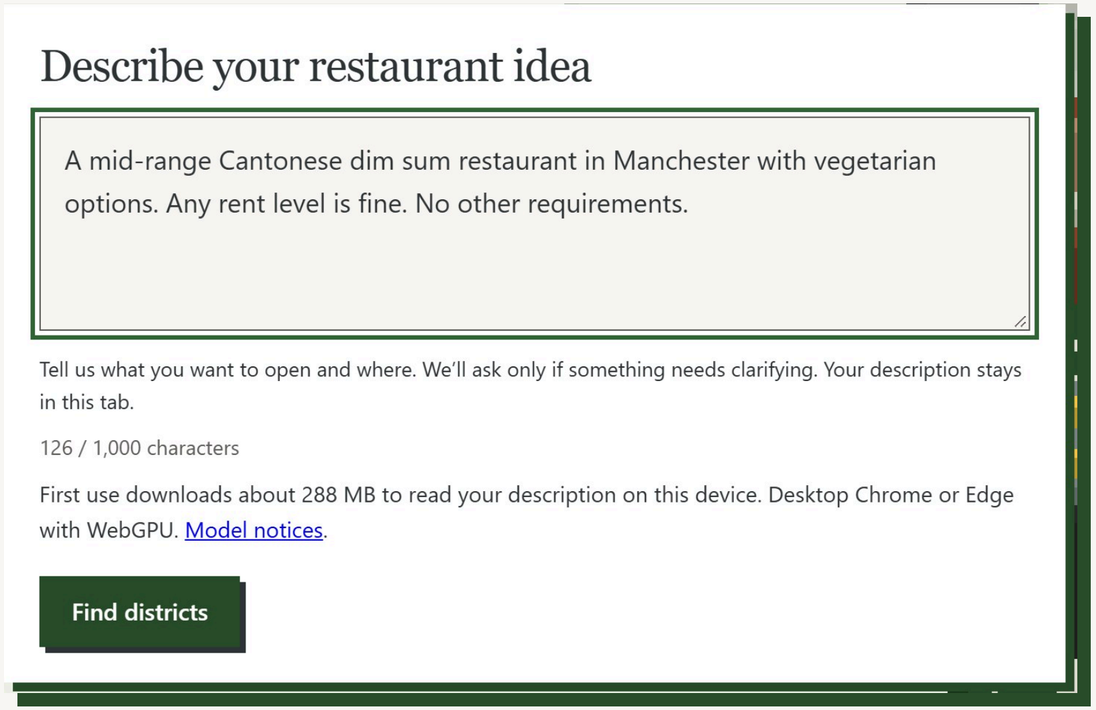
Explore

See districts on a map, move the score threshold and narrow down a ranked shortlist.

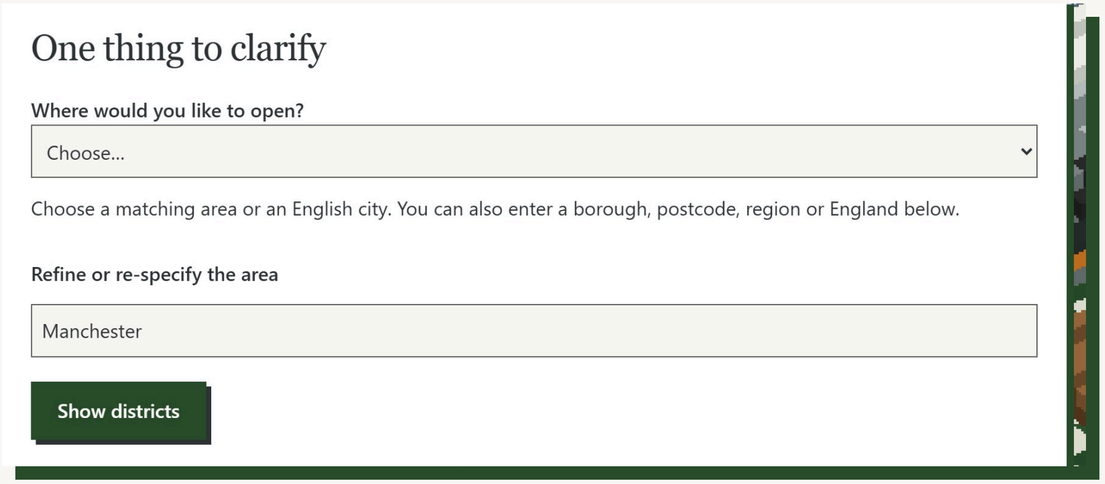
Each district gets a **review-based success score** from 0 to 1: how well restaurants like yours have done with diners there, judged by ratings and review activity.

The answer-reading AI is experimental. An editable review card is planned but not yet wired in.

Founders write it the way they would say it

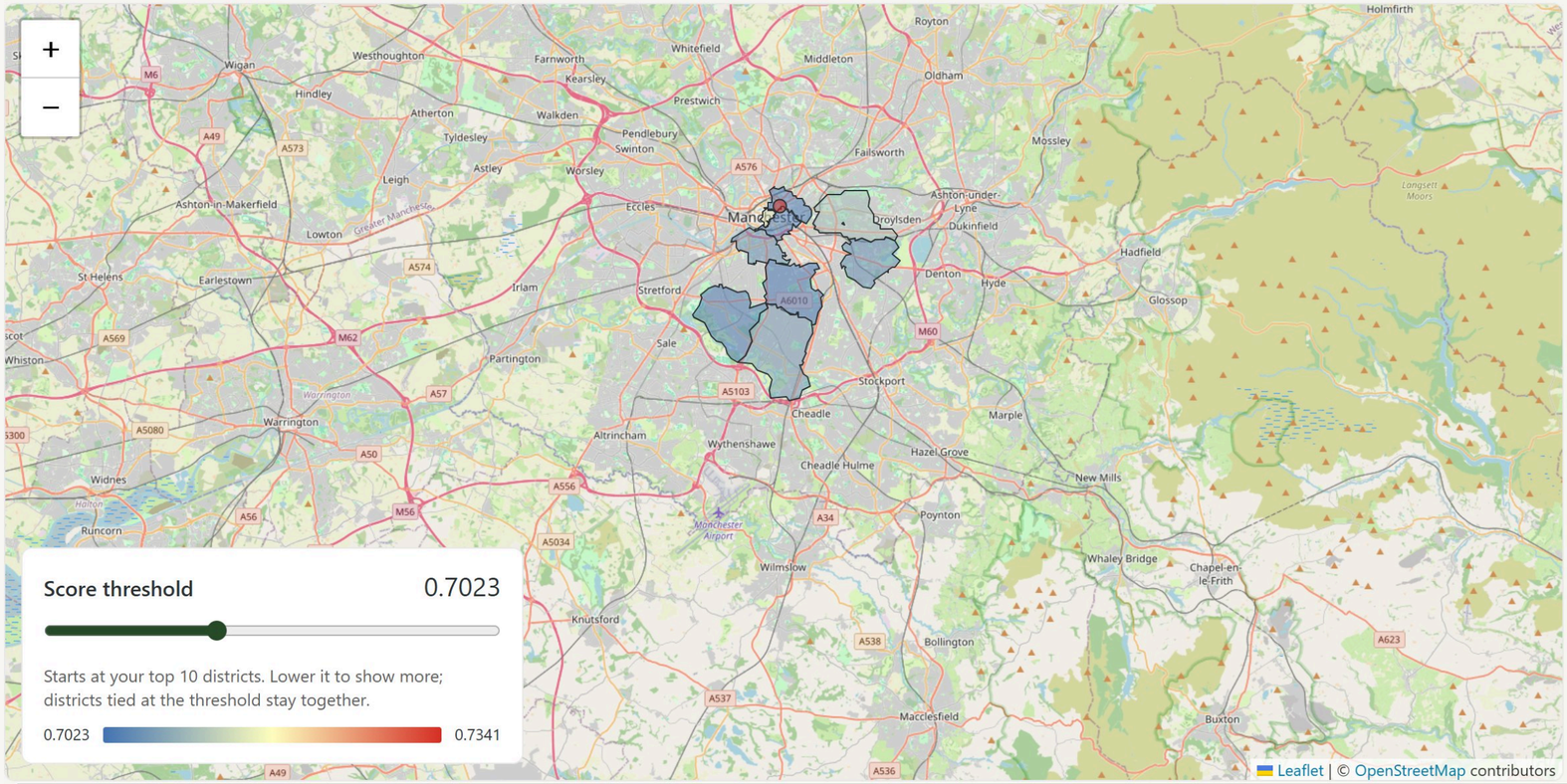


The description covers cuisine, prices, place, rent and diet in one sentence. The language model reading it runs on the user's own device, so the text never leaves the browser tab.



"Manchester" can mean the council area or the built-up town, so Plotluck asks one question instead of guessing. Everything else it read is kept.

Every district in the area, scored and ranked



District shortlist

10 of 19 districts in your search area

1	**M60**	0.7341
2	**M2**	0.7165
3	**M11**	0.7108
4	**M20**	0.7071
5	**M15**	0.7053
6	**M18**	0.7051
7	**M1**	0.7031
8	**M21**	0.7030
9	**M4**	0.7026
10	**M14**	0.7023

Read for you. “Cantonese dim sum” became the Chinese cuisine, with mid-range prices, vegetarian options and any rent. Inferred values are labelled as inferred.

Scored by the model. All 19 Manchester districts were scored. The threshold starts at the top 10; lower it to see more.

Honest on screen. The page states that the score is not a chance of survival or profit, and that closed restaurants may be missing from the data.

A language model reads, a trained network scores

Read

A small open language model turns the description into six fields: area, cuisine, menu prices, rent, restaurant type and other requirements.

NN2 · on the user’s device

Check

Every field must quote the description as evidence. Places are matched to an offline map of England. Anything missing or unclear becomes a question.

Validator · rules, not AI

Score

A neural network trained on 130,337 English restaurants scores the profile in every district of the search area.

NN1 · trained on IBM LinuxONE

Everything runs in the browser from a static GitHub Pages site. There is no server to pay for or switch off, so Plotluck stays live after the event.

No user data is collected. The description is held only in the open tab.

Two models, one answer

In
“A mid-range Cantonese dim sum restaurant in Manchester…”

→

NN2 reads

Qwen2.5-0.5B-Instruct, open weights

- Runs in the browser with WebLLM on WebGPU; about 288 MB, downloaded once
- Reads up to 1,000 characters
- Returns six fields as JSON, each with the quote it came from
- Marks every value as stated or inferred

→

Validator

Deterministic rules

- Rejects values not grounded in the text
- Resolves places offline
- Asks only for what is missing

→

NN1 scores

Neural network, 94 → 64 → 32 → 1

- 8 restaurant inputs: cuisine, cuisine group, type, price band, chain and three dietary flags
- 15 district inputs: population, age, ethnicity, sex, deprivation, rent, restaurant supply, attractions, hotels
- ReLU layers, sigmoid output: a 0–1 score per district

Why a small language model? It is small enough to run on an ordinary laptop, so a founder’s plans never leave their machine.

Why a separate scorer? Scoring needs numbers learned from real outcomes. NN1 is tiny, fast and exported as plain JSON weights with a JavaScript forward pass.

130,337 restaurants, tested on districts the model never saw

144,681

restaurant listings in England



142,107

after cleaning



Removed 1,177 without a location, 620 duplicate listings, 70 far from any postcode and 707 in districts that cross the Welsh or Scottish border.

130,337

with a usable rating label



The label combines average rating, the mix of review grades and review volume, damped by 20 prior reviews so a handful of reviews cannot dominate.

How we tested it

Five-fold cross-validation with whole postcode districts held out, so the test always uses districts the model never trained on. Each fold keeps its own early-stopping slice and is compared with a baseline that always predicts the average.

District context, all open data: Census 2021 (population, age, ethnicity, sex), Indices of Deprivation 2025, Valuation Office Agency rent, OpenStreetMap attractions and hotels, and the ONS Postcode Directory.

Run results: runs/2026-10-17-train-nn1-context/metrics.json. Restaurant data: a 2021 Tripadvisor-derived dataset published on Kaggle.

We trained the model on IBM LinuxONE

```
linux1@adriel:~$ hostname; lscpu | head -20
adriel
Architecture:                         s390x
CPU op-mode(s):                       32-bit, 64-bit
Byte Order:                           Big Endian
CPU(s):                               3
On-line CPU(s) list:                  0-2
Vendor ID:                            IBM/S390
Model name:                           -
Machine type:                         9175
Thread(s) per core:                   1
Core(s) per socket:                   1
Socket(s) per book:                   1
Book(s) per drawer:                   1
Drawer(s):                            3
CPU dynamic MHz:                      5508
CPU static MHz:                       5508
BogoMIPS:                             28901.00
Dispatching mode:                     horizontal
Flags:                                esan3 zarch stfle msa ldisp eimm dfp edat etf3eh highgprs te vx vxd vxe gs vxe2
 vxp sort dflt vxp2 nnpa sie
Hypervisor:                           z/VM 7.3.0
Hypervisor vendor:                    IBM
```

Our LinuxONE virtual machine: **s390x, big-endian**, IBM machine type 9175 on z/VM 7.3. From docs/evidence/01-ssh-lscpu.png.

- **Ingest and clean** the full restaurant and district data in Jupyter on the VM.
- **Train and cross-validate NN1** on LinuxONE. The Tripadvisor-derived rows never leave the server.
- **Export the weights** as JSON so the website can keep scoring after the VM expires.
- **Check byte order:** weights written on a big-endian machine are load-tested on x86 before release.

More evidence in the repo: docs/evidence/03-jupyterlab.png, 04-ingest.png, 05-train-nn1.png, 09-weight-export.png.

39% lower error than always guessing the average

0.0503

NN1 mean absolute error

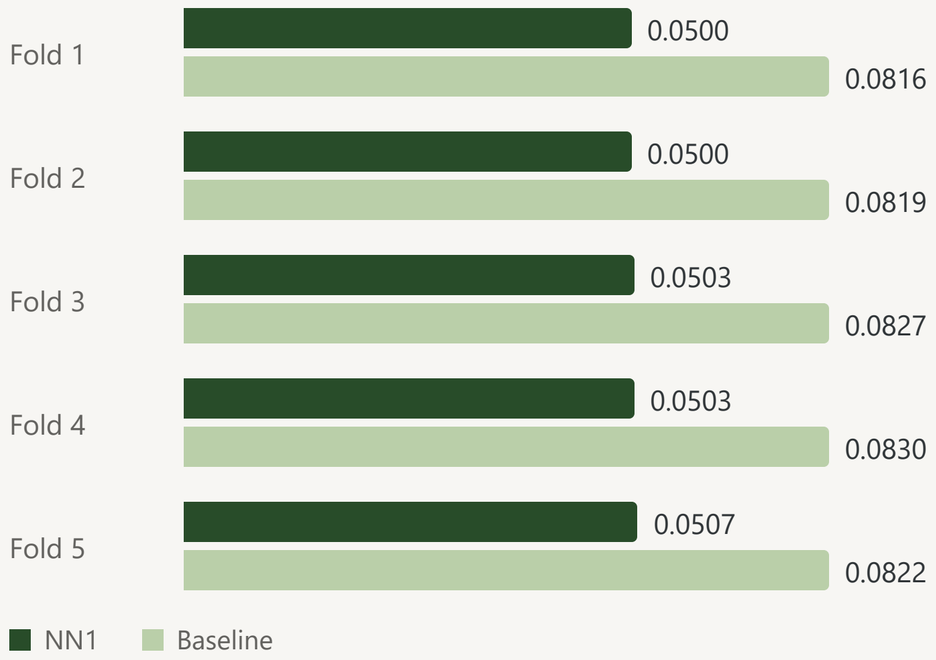
0.0823

Baseline mean absolute error

Average prediction error on the 0–1 score across five district-grouped test folds; lower is better. About 26,000 restaurants are tested in each fold.

The score uses customer ratings and review activity. It does not predict profit or whether a restaurant will survive. We still need to check how well the district ranking works for each cuisine.

Every fold, same result



Source: runs/2026-10-17-train-nn1-context/metrics.json (the model the live site uses).

A shortlist, not a promise

Data stays where it belongs

Tripadvisor-derived training rows stay on LinuxONE. We publish only model outputs, with attribution, for non-commercial use. Restaurant samples in the repo are synthetic.

Private by design

The description is read on the user's device and held only in the open tab. Nothing is sent to a server or stored.

Asks instead of guessing

Each field must be backed by the user's own words. Inferred values are marked, and missing or unclear details become a question.

Limits, stated plainly

The data may miss restaurants that closed. District averages cannot tell you what one street or lease will be like. Founders should visit sites and check the numbers themselves. The answer-reading AI and LinuxONE acceleration benchmark are still being tested.

Know where to look first

	Gut feel and site visits	Paid location research	Plotluck
Cost	Free, but slow	Paid, per study	**Free**
Areas compared	A handful	The ones in the brief	**Every district in the area**
Fits your cuisine	Depends on who you ask	Usually	**Yes, 115 cuisines**
Your plans stay private	Yes	Shared with a consultant	**Yes, read on your device**

Next

- Validate the district rankings for each cuisine
- Finish testing the answer-reading AI
- Wire in the editable review card

Live demo
adrielau.github.io/PlotLuck

Code and evidence
github.com/AdrielAu/PlotLuck

Plotluck gives independent founders a clearer starting point: where to start visiting streets and checking costs.